

NALLABATI VENKATA DURGA SAI SARAT CHANDRA

Target Role: Data Analyst Intern • Swiggy

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EDUCATION & NATIONAL MERIT PEDIGREE

Amrita Vishwa Vidyapeetham

August 2024 – Expected May 2028

B.Tech in Robotics & Artificial Intelligence (Cumulative GPA: 9.05 / 10.00, Top 5% Cohort)

Amritapuri, India

- **Academic Standing:** AP POLYCET State Merit Rank: 497 (Top 0.5% Statewide Percentile).
- **National Distinctions:** ISRO-IIRS YUVIKA (Selected for space science analytics) | NAEST National Physics Test (Cleared Prelims & Finals, IAPT / Prof. H.C. Verma) | MTSO Olympiad (Level 2 Qualifier).

PEER-REVIEWED PUBLICATIONS & PROCEEDINGS

- N. V. D. S. Sarat Chandra, M. Nunna, Asha C. A., P. C. Sreekumar, “Small Scale Raspberry Pi Based Autonomous Car: A Practical Guide to Understand Complexity.” in *Proc. IEEE ICRM*, 2025.
- N. V. D. S. Sarat Chandra, et al., “Physics-Guided Diffusion with Thermodynamic Awareness for 3D Molecular Generation.” Submitted to *IEEE IES*, 2026.
- N. V. D. S. Sarat Chandra, et al., “Domain-Aware Multi-Modal Industrial Diagnostics: Integrating Mechanical Heuristics with Conformal Uncertainty.” Submitted 2026.

TARGETED ENGINEERING PROJECTS & SYSTEMS ARCHITECTURE

Blood Neural Network: PINNs for Inverse Hemodynamic Rheology Discovery

SciML Researcher

PINNs, Navier-Stokes, Carreau-Yasuda, ANSYS Fluent CFD, NTK Trace Balancing

- **Core Impact:** <2.4% relative L2 error vs high-resolution ANSYS Fluent pulsatile CFD benchmarks.

Domain-Aware Industrial Failure Prediction & Diagnostics Framework

PIML Researcher / Data Science Lead

Pandas, NumPy, Optuna Bayesian Ensemble, Conformal Uncertainty, XGBoost

- **Core Impact:** 85.2% recall at 91.4% precision (0.94 AUROC) across 120k telemetry time-series samples.

Multi-Parameter Health Vitals Monitor & Real-Time IoT Data Streamer

Biomedical Systems Developer

MAX30102 PPG, ESP8266/ESP32, Fast-Mode I2C, 2nd-order IIR Bandpass, FreeRTOS

- **Core Impact:** <200ms real-time SpO2/HRV cloud telemetry; 9-clock I2C bus lockup recovery.

TECHNICAL SKILLS & CORE COMPETENCIES

Core Languages & Math: Python (3.12), SQL, C++, MATLAB, Multivariable Calculus, Linear Algebra, Probability

Data Science & ML Stack: Pandas, NumPy, Scikit-Learn, Optuna (Bayesian Optimization), PyTorch, Conformal Uncertainty

Data Pipelines & Time-Series: EDA, Time-Series Feature Engineering, ETL Ingestion, REST APIs, JSON/Parquet, SVD/POD

Analytics & Tooling: Jupyter, Git/GitHub, Docker, Linux (Ubuntu Server), ANSYS Fluent, LaTeX, Statistical Testing